

Identifying Risks in AI Healthcare Case Studies

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Introduction

Artificial intelligence (AI) refers to computer systems designed to perform tasks that typically require human intelligence, such as learning from data, recognizing patterns, and making decisions. In healthcare, AI is increasingly being used to improve diagnostic accuracy, enhance patient care, streamline administrative processes, and predict disease outcomes. AI systems can analyze medical images like X-rays and MRI scans, assist in drug discovery, manage patient data, and support clinical decision-making. While these applications offer tremendous benefits for patients and healthcare providers, they also introduce significant risks that healthcare organizations must carefully manage.

Three Major Risks of AI in Healthcare

Risk 1: Data Privacy and Security Breaches

What is this risk? AI systems in healthcare collect and store vast amounts of sensitive patient data, including medical histories, genetic information, and personal identifiers. This data becomes an attractive target for cybercriminals who may attempt to steal, alter, or misuse it. Healthcare providers face unique privacy attacks on AI algorithms, including membership inference attacks (where attackers try to determine if someone was in the training data), reconstruction attacks (where attackers try to recreate original data), and property inference attacks (where attackers learn about the algorithm's properties). Additionally, healthcare employees may unknowingly use unvetted AI tools like ChatGPT to handle sensitive patient information, creating unintended privacy risks (Moore, 2023; Clearwater Security, n.d.).

Example impact: A hospital implements an AI-powered patient management system that collects data on medication dosages, diagnoses, and treatment outcomes. Hackers breach the

system and steal the personal health information of 100,000 patients. Patients may experience identity theft, have their genetic information misused by insurance companies, or suffer emotional distress from knowing their private medical details were exposed. Healthcare providers could face lawsuits and regulatory penalties for failing to protect patient privacy.

Risk 2: Algorithmic Bias and Discrimination

What is this risk? AI systems trained on biased data may reflect and amplify existing inequalities in healthcare. If the data used to train an AI system underrepresents certain groups or contains historical biases, the algorithm may make discriminatory decisions. For example, an AI risk assessment tool that was trained on healthcare spending data may inadvertently learn to prioritize patients based on cost rather than clinical need, leading to biased treatment recommendations. These biases are particularly dangerous because AI outputs appear objective and unbiased on the surface, even though they are built into algorithms by humans (Rosenstrauch et al., 2024; Clearwater Security, n.d.).

Example impact: An AI diagnostic system trained primarily on imaging data from one racial group performs poorly when diagnosing diseases in patients from other racial backgrounds. This leads to delayed diagnoses and poorer health outcomes for minority patients. Similarly, an AI system that recommends organ transplant recipients based on training data reflecting systemic biases may consistently prioritize certain patients over others who might benefit more from the transplant, raising serious ethical and legal concerns.

Risk 3: Lack of Transparency and Accountability

What is this risk? Many AI systems operate as "black boxes," meaning the logic behind their decisions is unclear or difficult to explain. Healthcare providers and patients may not understand why an AI system recommended a particular diagnosis or treatment plan. This lack of

transparency makes it difficult for clinicians to validate AI recommendations, patients to give informed consent, and regulators to hold developers accountable when things go wrong. When errors occur or patients are harmed, it is often unclear who bears responsibility, the AI developer, the healthcare organization, or the healthcare provider. This ambiguity undermines trust and makes it difficult to establish liability and learn from mistakes (Rosenstrauch et al., 2024; Clearwater Security, n.d.).

Example impact: A cardiologist uses an AI system to evaluate patient chest X-rays and receives a recommendation to perform heart surgery on a patient. The AI system does not explain its reasoning, and the cardiologist cannot understand why it made this recommendation. If the surgery is unnecessary and harms the patient, it becomes difficult to determine fault. Was the AI system wrong? Did the cardiologist fail to critically evaluate the recommendation? This confusion can lead to litigation, loss of patient trust, and difficulties implementing necessary safeguards.

Deep Dive: Algorithmic Bias as a Critical Risk

Why is algorithmic bias particularly important? Among the three risks identified, algorithmic bias and discrimination stand out as particularly important for several reasons. First, it affects fundamental principles of healthcare, equity and justice. Healthcare is built on the premise that all patients deserve equal quality care regardless of their background. When AI systems perpetuate bias, they undermine this core principle and can cause serious harm to vulnerable populations who already face healthcare disparities. Second, bias is often hidden and difficult to detect. Unlike a security breach, which is usually discovered and reported, biased decisions can persist silently for years, affecting thousands of patients before being identified. Third, the consequences of bias extend beyond individual harm, they erode public trust in both healthcare

institutions and AI technology itself. If communities discover that AI systems are discriminating against them, they may refuse to use these technologies or seek care elsewhere, widening healthcare inequities.

How can healthcare organizations reduce this risk? Healthcare organizations can implement several strategies to mitigate algorithmic bias. First, they should collect diverse and representative training data that includes patients from different racial, ethnic, socioeconomic, and geographic backgrounds. Second, they should conduct regular bias audits and testing of AI systems to identify and measure any discriminatory patterns. Third, they should involve diverse teams in the design and development of AI systems, including clinicians, data scientists, ethicists, and patient representatives from different communities. Fourth, organizations should maintain transparency by documenting how AI systems are trained and what data is used. Finally, healthcare providers should validate AI recommendations against clinical judgment rather than accepting them uncritically, and patient information about how AI is being used in their care should be disclosed to enable informed decision-making (Rosenstrauch et al., 2024).

Conclusion: Do the Benefits Outweigh the Risks?

Based on the research reviewed, the benefits of AI in healthcare are substantial but the risks are real and cannot be ignored. AI has demonstrated remarkable potential to improve diagnostic accuracy, enable earlier disease detection, increase access to care in underserved areas, and reduce the burden on healthcare workers facing workforce shortages. These benefits could save lives and improve health outcomes for millions of people globally. However, the risks of data breaches, algorithmic bias, and lack of transparency are serious threats that could cause significant harm if not properly managed. Rather than viewing this as a simple choice between benefits and risks, the answer lies in responsible AI development and deployment. Healthcare

organizations, AI developers, regulators, and policymakers must work together to establish strong governance frameworks, implement robust security measures, conduct rigorous testing for bias, maintain transparency, and hold parties accountable when things go wrong. The question is not whether we should use AI in healthcare, but how we can do so safely and equitably. When appropriate safeguards are in place, the benefits of AI in healthcare can indeed outweigh the risks, leading to better patient outcomes and more efficient, equitable healthcare systems. However, these safeguards must be implemented proactively, not reactively after harm has occurred.

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